

Logical backup in multitenant database

PDB-to-PDB Migration Using Oracle Data Pump

In multitenant database, I've taken a schema level backup. Below are the following steps:

1. Create a directory in OS level.

```
[root@localhost ~]# mkdir -p /u02/backup
[root@localhost ~]# chown -R oracle:oinstall /u02
[root@localhost ~]# chmod -R 775 /u02
[oracle@localhost backup]$ pwd
/u02/backup
```

```
[root@localhost ~]# mkdir -p /u02/backup
[root@localhost ~]# chown -R oracle:oinstall /u02
[root@localhost ~]# chmod -R 775 /u02
```

2. Create a same directory in logical level

```
SQL> create directory exp_dir as '/u02/backup';
SQL> grant read, write on directory exp_dir to system;
Grant succeeded.
```

```
SQL> conn
Enter user-name: sys/oracle@pdb1 as sysdba
Connected.
SQL>
SQL>
SQL> show pdbs;

  CON_ID CON_NAME          OPEN MODE  RESTRICTED
-----
       3 PDB1              READ WRITE NO
```

```
SQL>
SQL> create directory exp_dir as '/u02/backup';
Directory created.
```

3. Give the permission of the directory.

```
SQL> grant read,write on directory exp_dir to system;
Grant succeeded.
```

```
SQL> grant read,write on directory exp_dir to system;
Grant succeeded.
```

4. Take the schema backup. (Schema name is scott)

expdp system/oracle@pdb1 directory=exp_dir dumpfile=scott.dmp logfile=scott.log
schemas=scott

```
[oracle@localhost backup]$ expdp system/oracle@pdb1 directory=exp_dir dumpfile=scott.dmp logfile=scott.log schema=scott
LRM-00101: unknown parameter name 'schema'

[oracle@localhost backup]$ expdp system/oracle@pdb1 directory=exp_dir dumpfile=scott.dmp logfile=scott.log schemas=scott

Export: Release 19.0.0.0.0 - Production on Tue Jul 28 15:37:04 2026
Version 19.3.0.0.0

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```

```
Connected to: Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Starting "SYSTEM"."SYS_EXPORT_SCHEMA_01": system/*****@pdb1 directory=exp_dir dumpfile=scott.dmp logfile=scott.log schemas=scott
Processing object type SCHEMA_EXPORT/USER
Processing object type SCHEMA_EXPORT/SYSTEM_GRANT
Processing object type SCHEMA_EXPORT/DEFAULT_ROLE
Processing object type SCHEMA_EXPORT/TABLESPACE_QUOTA
Processing object type SCHEMA_EXPORT/PRE_SCHEMA/PROCACT_SCHEMA
Master table "SYSTEM"."SYS_EXPORT_SCHEMA_01" successfully loaded/unloaded
*****
Dump file set for SYSTEM.SYS_EXPORT_SCHEMA_01 is:
/u02/backup/scott.dmp
Job "SYSTEM"."SYS_EXPORT_SCHEMA_01" successfully completed at Tue Jul 28 15:37:52 2026 elapsed 0 00:00:43
```

5. Now, remove the user from pdb database

SQL> drop user scott cascade;

User dropped.

```
SQL> drop user scott cascade;

User dropped.

SQL> col username for a10;
SQL>
SQL> select username from dba_users where username ='SCOTT';

no rows selected
```

6. Import the schema in pdb

[oracle@localhost backup]\$ impdp system/oracle@pdb1 directory=exp_dir
dumpfile=scott.dmp logfile=scott_imp.log

```
[oracle@localhost backup]$ impdp system/oracle@pdb1 directory=exp_dir dumpfile=scott.dmp logfile=scott_imp.log

Import: Release 19.0.0.0.0 - Production on Tue Jul 28 15:49:10 2026
Version 19.3.0.0.0

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```

```
Connected to: Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Master table "SYSTEM"."SYS_IMPORT_FULL_01" successfully loaded/unloaded
Starting "SYSTEM"."SYS_IMPORT_FULL_01": system/*****@pdb1 directory=exp_dir dumpfile=scott.dmp logfile=scott_imp.log
Processing object type SCHEMA_EXPORT/USER
Processing object type SCHEMA_EXPORT/SYSTEM_GRANT
Processing object type SCHEMA_EXPORT/DEFAULT_ROLE
Processing object type SCHEMA_EXPORT/TABLESPACE_QUOTA
Processing object type SCHEMA_EXPORT/PRE_SCHEMA/PROCACT_SCHEMA
Job "SYSTEM"."SYS_IMPORT_FULL_01" successfully completed at Tue Jul 28 15:49:25 2026 elapsed 0 00:00:11
```

7. Check the schema is coming or not after import the schema

SQL> select username from dba_users where username ='SCOTT';

```
[oracle@localhost backup]$ sqlplus
SQL*Plus: Release 19.0.0.0.0 - Production on Tue Jul 28 15:50:34 2026
Version 19.3.0.0.0

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Enter user-name: sys/oracle@pdb1 as sysdba

Connected to:
Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.3.0.0.0
```

```
SQL> show pdbs;
```

CON_ID	CON_NAME	OPEN MODE	RESTRICTED
3	PDB1	READ WRITE	NO

```
SQL> select username from dba_users where username ='SCOTT';
```

```
USERNAME
-----
SCOTT
```

8. Create a logical directory in pdb2 database.

SQL> create directory exp_dir_2 as '/u02/backup';

```
SQL> create directory exp_dir_2 as '/u02/backup';
Directory created.
```

9. Give the permission to directory.

SQL> grant read, write on directory exp_dir_2 to system;

Grant succeeded.

```
SQL> grant read, write on directory exp_dir_2 to system;
Grant succeeded.
```

10. Import the schema backup from pdb1 to pdb2

[oracle@localhost backup]\$ impdp system/oracle@pdb2 directory=exp_dir_2 dumpfile=scott.dmp logfile=scott_pdb2.log

```
[oracle@localhost backup]$ impdp system/oracle@pdb2 directory=exp_dir_2 dumpfile=scott.dmp logfile=scott_pdb2.log
Import: Release 19.0.0.0.0 - Production on Tue Jul 28 16:06:59 2026
Version 19.3.0.0.0

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```

```
Connected to: Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Master table "SYSTEM"."SYS_IMPORT_FULL_01" successfully loaded/unloaded
Starting "SYSTEM"."SYS_IMPORT_FULL_01": system/*****@pdb2 directory=exp_dir_2 dumpfile=scott.dmp logfile=scott_pdb2.log
Processing object type SCHEMA_EXPORT/USER
Processing object type SCHEMA_EXPORT/SYSTEM_GRANT
Processing object type SCHEMA_EXPORT/DEFAULT_ROLE
Processing object type SCHEMA_EXPORT/TABLESPACE_QUOTA
Processing object type SCHEMA_EXPORT/PRE_SCHEMA/PROCACT_SCHEMA
Job "SYSTEM"."SYS_IMPORT_FULL_01" successfully completed at Tue Jul 28 16:07:09 2026 elapsed 0 00:00:08
```

11. Check the schema in pdb2

Enter user-name: sys/oracle@pdb2 as sysdba

Connected to:

Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production

Version 19.3.0.0.0

SQL> select username from dba_users where username='SCOTT';

USERNAME

SCOTT

```
[oracle@localhost backup]$ sqlplus

SQL*Plus: Release 19.0.0.0.0 - Production on Tue Jul 28 16:08:28 2026
Version 19.3.0.0.0

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Enter user-name: sys/oracle@pdb2 as sysdba

Connected to:
Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.3.0.0.0
```

```
SQL>
SQL> select username from dba_users where username='SCOTT';

USERNAME
-----
SCOTT
```